Field Service Management System - Database Design

Code First Database Design (SQL Server + ASP.NET Core Web API)

Users

Column	SQL Type
Id	int
FirstName	nvarchar(100)
LastName	nvarchar(100)
Email	nvarchar(150)
PasswordHash	nvarchar(max)
PhoneNumber	nvarchar(15)
Role	nvarchar(30)
ProfileImage	nvarchar(max)
IsActive	bit
CreatedAt	datetime2
UpdatedAt	datetime2

CustomerAddresses

Column	SQL Type
Id	int
CustomerId	int
AddressTitle	nvarchar(50)
HouseNumber	nvarchar(50)
Street	nvarchar(200)
Area	nvarchar(200)
City	nvarchar(100)
State	nvarchar(100)
Pincode	nvarchar(10)
Latitude	decimal(10,8)
Longitude	decimal(11,8)
IsDefault	bit

Services

Column	SQL Type

Id	int
ServiceName	nvarchar(150)
Description	nvarchar(max)
Category	nvarchar(100)
BasePrice	decimal(10,2)
EstimatedHours	decimal(5,2)
Image	nvarchar(max)
IsActive	bit
CreatedAt	datetime2

Technicians

Column	SQL Type
Id	int
UserId	int
EmployeeCode	nvarchar(30)
Specialization	nvarchar(100)
Experience	int
Rating	decimal(3,2)
EmploymentStatus	nvarchar(30)
CurrentStatus	nvarchar(30)
JoiningDate	date
IsAvailable	bit

ServiceRequests

Column	SQL Type
Id	int
RequestNumber	nvarchar(30)
CustomerId	int
AddressId	int
ServiceId	int
TechnicianId	int
ProblemDescription	nvarchar(max)
PreferredDate	date
PreferredTime	time
Priority	nvarchar(20)
Status	nvarchar(30)
ManagerNotes	nvarchar(max)

EstimatedCost	decimal(10,2)
CreatedAt	datetime2
UpdatedAt	datetime2

WorkReports

Column	SQL Type
Id	int
ServiceRequestId	int
TechnicianId	int
Diagnosis	nvarchar(max)
WorkPerformed	nvarchar(max)
LabourHours	decimal(5,2)
LabourCharge	decimal(10,2)
Resolution	nvarchar(max)
TechnicianNotes	nvarchar(max)
WarrantyDays	int
ReviewStatus	nvarchar(30)
SubmittedAt	datetime2

WorkReportParts

Column	SQL Type
Id	int
WorkReportId	int
PartName	nvarchar(150)
Quantity	int
UnitPrice	decimal(10,2)
TotalPrice	decimal(10,2)

Invoices

Column	SQL Type
Id	int
InvoiceNumber	nvarchar(30)
ServiceRequestId	int
CustomerId	int
GeneratedBy	int
ServiceCharge	decimal(10,2)
LabourCharge	decimal(10,2)

PartsAmount	decimal(10,2)
GST	decimal(10,2)
Discount	decimal(10,2)
TotalAmount	decimal(10,2)
Status	nvarchar(20)
DueDate	date
PaidDate	datetime2
Notes	nvarchar(max)
CreatedAt	datetime2

InvoiceItems

Column	SQL Type
Id	int
InvoiceId	int
ItemName	nvarchar(150)
Quantity	int
UnitPrice	decimal(10,2)
TotalPrice	decimal(10,2)

Notifications

Column	SQL Type
Id	int
UserId	int
Title	nvarchar(150)
Message	nvarchar(max)
Type	nvarchar(50)
IsRead	bit
CreatedAt	datetime2

StatusHistories

Column	SQL Type
Id	int
ServiceRequestId	int
Status	nvarchar(30)
ChangedByUserId	int
Notes	nvarchar(max)
ChangedAt	datetime2

Settings

Column	SQL Type
Id	int
UserId	int
Theme	nvarchar(20)
Language	nvarchar(20)
EmailNotification	bit
PushNotification	bit
DarkMode	bit

RefreshTokens

Column	SQL Type
Id	int
UserId	int
Token	nvarchar(max)
ExpiryDate	datetime2
CreatedAt	datetime2
Revoked	bit

Workflow: Customer → Manager Review → Technician → Work Report → Manager Approval → Invoice → Payment → Close Request

Main Relationships: Users → Technicians, Users → CustomerAddresses, ServiceRequests → WorkReports → WorkReportParts, ServiceRequests → Invoices → InvoiceItems, Users → Notifications, Users → Settings, Users → RefreshTokens.